PARISHRAM 2.0

CLASS 12TH 2026

Botany-Sexual Reproduction in Flowering Plants

BIOLOGY

LECTURE-5

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TOPICS TO BE COVERED

1. Pollination
2. Agents of pollination
3. Pollen Pistil Interaction And Biotic Agents
4.

Pollination

Devices / Agents that promote autogamy / self-pollination



1. Bisexuality

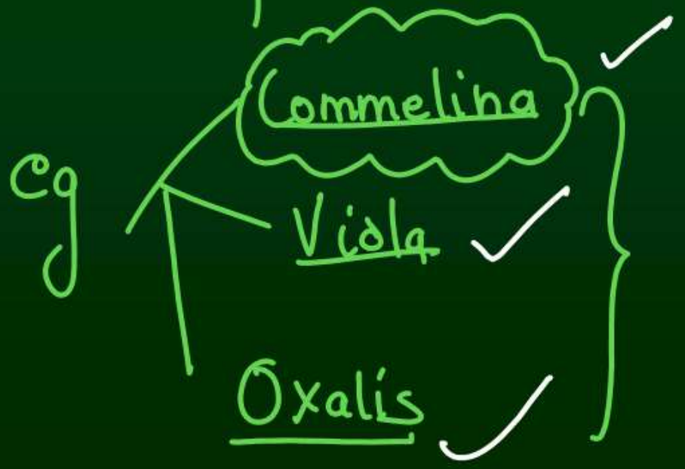
Bisexuality is a prime requirement for autogamy but does not gaurantee it

2. Both the sex organ in a ♂ flower should be kept close to each other & there should be no barrier in between them.

3. Both the sex organ should mature at the same time (synchrony) (Homogamy)

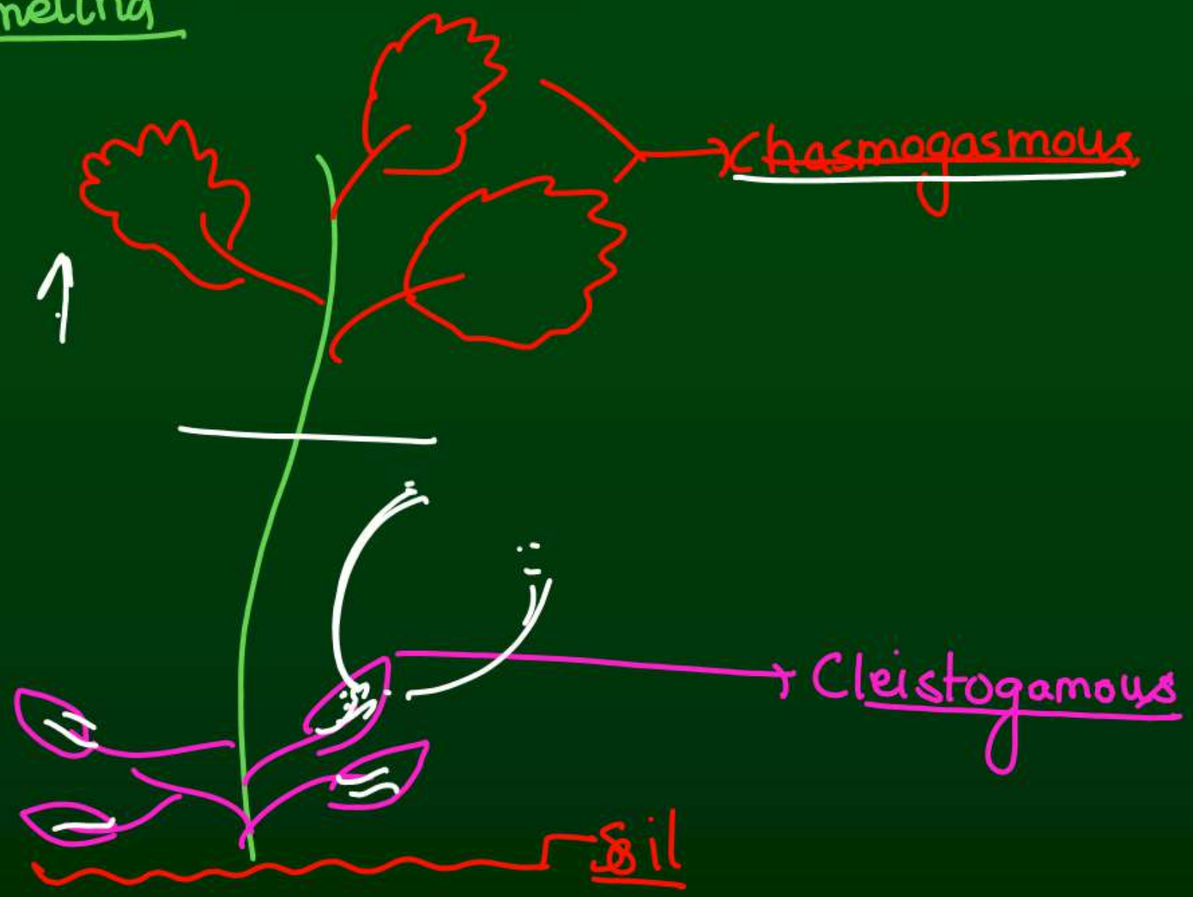
Even if all the conditions are fulfilled, still there is a scope of cross-pollination as the flower is open [chasmogamous].

(d) Cleistogamy → flowers which never opens such that no scope of cross pollination. Hence it guarantees self pollination → fertilization → seed formation.



Out-breeding devices

Groundling



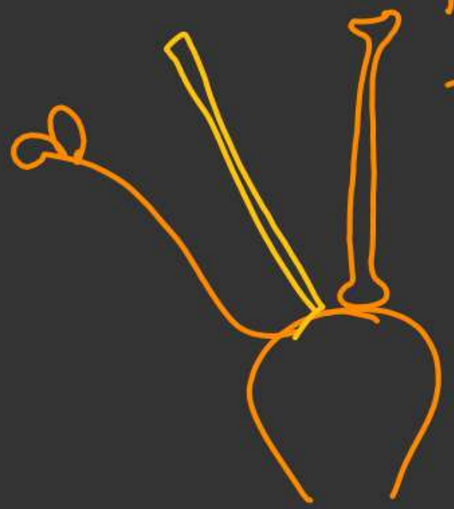
(i) Unisexual – Either a flower has ♂ sex organ or ♀ sex organ but not both

Hence there is no scope of self-pollination

(a) Monoecious → Geitonogamy

(b) Dioecious → Xenogamy

2 If a flower is ♀ and still wants cross pollination



then both the sex organs should be kept apart ✓

+

→ Presence of barrier in btw the sex organ

→

→ There should be no synchrony in the maturation of both the sex organ.

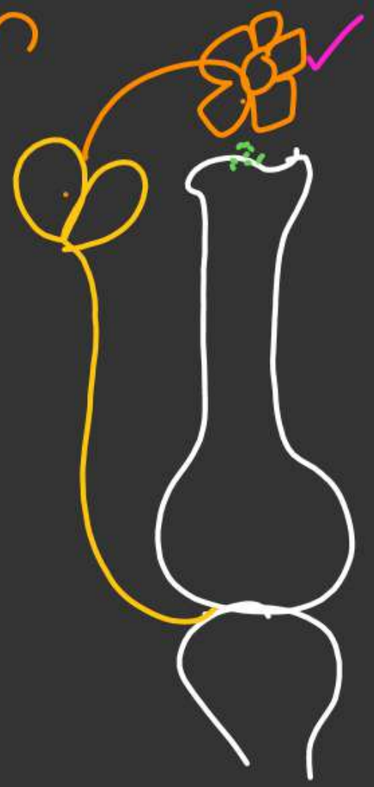
(Heterogamy)



Self-incompatibility

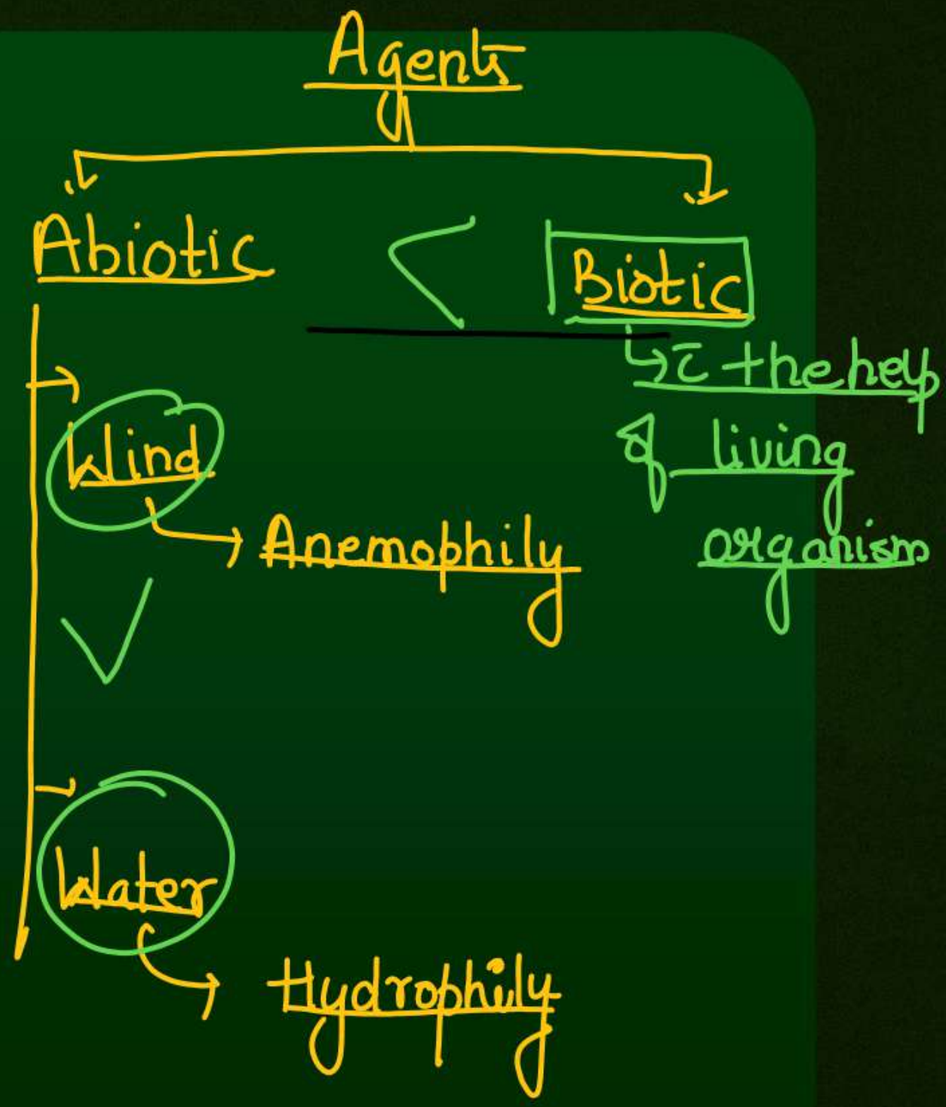
It is a genetical checkup where self-pollen fall on it own stigma, then

Pistil has the power to reject the germination of pollen on grounds of incompatibility



As soon as PG lands on the pistil, Pistil checks the PG for compatibility by releasing chemical

Agents of Pollination : Plants use two abiotic (wind and water) and one biotic (animals) agents to achieve pollination. Majority of plants use biotic agents for pollination. Only a small proportion of plants use abiotic agents. Pollen grains coming in contact with the stigma is a chance factor in both wind and water pollination. To compensate for this uncertainties and associated loss of pollen grains, the flowers produce enormous amount of pollen when compared to the number of ovules available for pollination.



Pollination by wind is more common amongst abiotic pollinations. Wind pollination also requires that the pollen grains are light and non-sticky so that they can be transported in wind currents. They often possess well-exposed stamens (so that the pollens are easily dispersed into wind currents, Figure 1.10) and large often-feathery stigma to easily trap air-borne pollen grains. Wind-pollinated flowers often have a single ovule in each ovary and numerous flowers packed into an inflorescence; a familiar example is the corn cob – the ears you see are nothing but the stigma and style which wave in the wind to trap pollen grains. Wind-pollination is quite common in grasses.

Note:—

Since abiotic agents donot confirm high rate of pollination success

∴

Majority of flowering plants use a range of animals as pollinating agents. Bees, butterflies, flies, beetles, wasps, ants, moths, birds (sunbirds and humming birds) and bats are the common pollinating agents. (Figure 1.11b). Among the animals, insects, particularly bees are the dominant biotic pollinating agents. Even larger animals such as some primates (lemurs), arboreal (tree-dwelling) rodents, or even reptiles (gecko lizard and garden lizard) have also been reported as pollinators in some species.

Often flowers of animal-pollinated plants are specifically adapted for a particular species of animal.

Majority of insect-pollinated flowers are large, colourful, fragrant and rich in nectar. When the flowers are small, a number of flowers are clustered into an inflorescence to make them conspicuous. Animals are attracted to flowers by colour and/or fragrance. The flowers pollinated by flies and beetles secrete foul odours to attract these animals. To sustain animal visits, the flowers have to provide rewards to the animals. Nectar and pollen grains are the usual floral rewards. For harvesting the reward(s) from the flower the animal visitor comes in contact with the anthers and the stigma. The body of the animal gets a coating of pollen grains, which are

generally sticky in animal pollinated flowers. When the animal carrying pollen on its body comes in contact with the stigma, it brings about pollination.

In some species floral rewards are in providing safe places to lay eggs; an example is that of the tallest flower of *Amorphophallus* (the flower itself is about 6 feet in height). A similar relationship exists between a species of moth and the plant *Yucca* where both species – moth and the

plant – cannot complete their life cycles without each other. The moth deposits its eggs in the locule of the ovary and the flower, in turn, gets pollinated by the moth. The larvae of the moth come out of the eggs as the seeds start developing.

Outbreeding Devices: Majority of flowering plants produce hermaphrodite flowers and pollen grains are likely to come in contact with the stigma of the same flower. Continued self-pollination result in inbreeding depression. Flowering plants have developed many devices to discourage self-pollination and to encourage cross-pollination. In some species, pollen release and stigma receptivity are not synchronised. Either the pollen is released before the stigma becomes receptive or stigma becomes receptive much before the release of pollen. In some other species, the anther and stigma are placed at different positions so that the pollen cannot come in contact with the stigma of the same flower. Both these devices prevent autogamy. The third device to prevent inbreeding is self-incompatibility. This is a genetic mechanism and prevents self-pollen (from the same flower or other flowers of the same plant) from fertilising the ovules by inhibiting pollen germination or pollen tube growth in the pistil. Another device to prevent self-pollination is the production of unisexual flowers. If both male and female flowers are present on the same plant such as castor and maize (monoecious), it prevents autogamy but not geitonogamy. In several species such as papaya, male and female flowers are present on different plants, that is each plant is either male or female (dioecy). This condition prevents both autogamy and geitonogamy.

QUESTION

Polygonum type of embryo sac is

1. 8-celled
2. 15-nucleated
3. Haploid
4. Exosporic type



QUESTION

Mark the odd option (w.r.t. contrivances of autogamy)

1. Homogamy ✓
2. Cleistogamy ✓
3. Dicliny — Unisexuality → cross polli
4. Bud pollination ✓



QUESTION

Which of the following pollination is common amongst abiotic agents?

1. Hydrophily
2. Entomophily
3. Ornithophily
4. Anemophily



QUESTION

Epihydrophily is found in

1. Tape grass
2. Sea grass
3. Lotus
4. Alisma



QUESTION

Which of the following plant provide safe place to insect for laying eggs?

1. Sage plant
2. Ophrys
3. Centaurea
4. Amorphophallus



QUESTION

Pollination occurs by pseudocopulation mechanism in

1. Ophrys
2. Fig
3. Mango
4. Zea may



THANK YOU